RESPONSIBLE DATA CARD

Customer Churn Prediction Dataset

Internship: Artificial Intelligence & Machine Learning

Organization: RabTech Academy

Project: Responsible Customer Churn Prediction

1. Dataset purpose

The dataset is intended to support exploratory customer churn analysis by identifying patterns associated with customers who have stopped using a service.

It may support analysis and a prototype retention-support workflow. It must not independently determine account cancellation, service denial, pricing, eligibility, or other consequential customer decisions.

2. Provenance and permission

The dataset was supplied for the RabTech Academy AI/ML internship task. The supplied files do not state the original data-collection process, source organization, data subjects, consent status, licence, or redistribution permissions.

These details must be confirmed with the dataset provider before using the data beyond this educational task or sharing it publicly. Do not assume the data is synthetic, anonymized, or licensed for redistribution.

3. Population and representation

The supplied CSV contains 12 customer records: 7 labelled 0 (not churned) and 5 labelled 1 (churned), an observed churn proportion of approximately 41.7%.

The plan_type column contains Basic, Standard, and Pro categories. The sample is too small to establish population representativeness or reliable group-level conclusions. Demographic attributes and information about missing or under-represented groups are not provided.

4. Features and target

customer_id: customer identifier; exclude it from model features because it is an identifier rather than a meaningful behaviour measure.

tenure_months: customer tenure in months.

support_tickets: number of support tickets.

monthly_spend_inr: monthly spend in INR.

last_login_days: days since last login.

plan_type: subscription plan category (Basic, Standard, or Pro).

churned: target label; 0 means not churned and 1 means churned.

5. Quality checks

Initial inspection found 12 rows and 7 columns, with no missing values and no duplicate rows. The target distribution is 7 non-churned and 5 churned records.

These checks do not establish that values are accurate, valid, current, or representative. Range checks, category validation, outlier review, and confirmation of observation dates still need to be completed.

Potential leakage: confirm that every feature was recorded before the churn outcome or prediction point. Do not use information created after churn as an input. Keep preprocessing inside the cross-validation pipeline and keep the target separate from features.

6. Risks and safeguards

Small-sample risk: estimates may be unstable. Treat results as a learning demonstration, not production evidence; obtain a larger, representative dataset before deployment.

Privacy and security: avoid publishing identifiable or confidential data; restrict access and confirm permission and applicable consent requirements.

Bias and representation: inspect errors across plan types only when enough data is available; do not claim fairness from tiny groups.

False positives: a non-churned customer may be incorrectly flagged. Require human review and avoid unnecessary or intrusive retention contact.

False negatives: a churned customer may be missed. Monitor missed cases and assess business costs.

Misuse: do not automatically cancel accounts, change prices, or deny services based on predictions. Use predictions only as decision support with human oversight.

7. Intended evaluation

Baseline: majority-class DummyClassifier.

Validation: Leave-One-Out Cross-Validation for this tiny educational dataset. Each fold trains on 11 records and tests on one; results remain highly uncertain.

Performance measures: accuracy, precision, recall, F1-score, and confusion matrix. Review false positives and false negatives separately.

Calibration: not meaningfully assessed with only 12 records; calibration requires more representative data.

Fairness: group-level evaluation by plan type is exploratory only and not reliable at this sample size.

Before real-world use, obtain more data, confirm permissions and provenance, use a time-aware evaluation design where appropriate, assess calibration and fairness with adequate group sizes, and monitor performance.

Dataset summary

Check	Initial result
Rows / columns	12 / 7
Missing values	0
Duplicate rows	0
Not churned (0)	7

Churned (1)	5
Observed churn proportion	41.7% (5 of 12)

Important limitation: The supplied materials do not document original provenance, consent, licensing, or representativeness. These require confirmation from the dataset provider.